



NCA Sponsored Training

Day 1 – Afternoon Session (Lab)

Basic secure communication tools

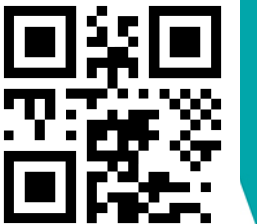
Presenter Name

City

Date



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Outline



1. Basic Cyphers

- *Cipher Wheel*
- *Phone Pad Cipher*

2. OpenSSL

- *Prime numbers*
- *AES Encryption*

2. Steganography

- *Steghide*
- *StegCracker*



Caesar Cipher



Visit the following URL:

<https://www.101computing.net/cipher-wheel.html>

Are you able to crack the following ciphertext?

vtcxjh lxiwdji tsjrpixdc xh axzt hxaktg xc iwt bxct



Phone Pad Cipher



Are you able to crack the following ciphertext?

55 2 88 7777 8

8 777 2 444 66 444
66 4

1	2 ABC	3 DEF
4 GHI	5 JKL	6 MNO
7 PQRS	8 TUV	9 WXYZ
*	0	#



OpenSSL – Symmetric Encryption



Check your OpenSSL version:

\$ openssl version

Check OpenSSL help:

\$ openssl help

- How many encryption methods are supported?



OpenSSL – Symmetric Encryption



Prime number test:

\$ openssl prime (number)

Are the following numbers prime?

- 42
- 1421
- 13137
- 997



OpenSSL – Symmetric Encryption



Create a file named myfile.txt with some text inside.

Encrypt with aes-256-cbc

```
$ openssl enc -aes-256-cbc -in myfile.txt -out encrypted.bin
```

Use the following command to view the output file: **cat encrypted.bin**

Is it easy to write out or transmit the output?





OpenSSL – Symmetric Encryption



Now repeat the previous command and add the **–base64** option.

```
$ openssl enc -aes-256-cbc -in myfile.txt -out encrypted.bin –base64
```

Use the following command to view the output file: **cat encrypted.bin**

Is it easy to write out or transmit the output?





OpenSSL – Symmetric Encryption



Now repeat the previous command and observe the encrypted output (use the same password)

```
$ openssl enc -aes-256-cbc -in myfile.txt -out encrypted.bin -base64
```

- Has the output changed?
- Why has it changed?



OpenSSL – Symmetric Encryption



Now let's decrypt the encrypted file:

```
$ openssl enc -d -aes-256-cbc -in encrypted.bin  
-pass pass:password -base64
```

- Has the output been decrypted correctly?
- What happens when you use the wrong password?



Steghide is a steganography program able to hide data in various kinds of

- Image
- audio-files

The color and the sample-frequencies, respectively, are not changed thus making the embedding resistant against first-order statistical tests.



Steganography



First, let's install the tool. In your **Kali Linux** VM, run the following commands:

```
$ sudo apt update
```

```
$ sudo apt install steghide
```

Have a look at the help:

```
$ steghide help
```



Steganography



Create a folder on your Kali Linux VM with an image file and a secret text file.

Make a copy of the image. You will use it to compare the original image with the stego file.

To hide the text in image, type

```
$ steghide embed -cf image.jpeg -ef secret-text.txt
```



Is there any difference between the original file and the one with your secret embedded?



Steganography



View information of embedded data

\$ steghide info image.jpeg

After you have embedded your secret data, the receiver has to use steghide in the following way:

steghide extract -sf image.jpeg



Stegcracker is a steganography brute-force utility to uncover hidden data inside files.

Using **stegcracker** is simple:

- pass a file to it as its first parameter and optionally pass the path to a wordlist of passwords to try as its second parameter.
- If you don't specify the wordlist, the tool will try to use the built-in rockyou.txt wordlist which ships with Kali Linux.



Steganalysis



First, let us install the tool. In your **Kali Linux** VM, run the following commands:

```
$ sudo apt update
```

```
$ sudo apt install stegcracker
```

Have a look at the help:

```
$ stegcracker --help
```



Steganalysis



Download a new image and embed a text file into it following the instructions below:

- Do not encrypt the file: Add the “**-e none**” option to the steghide command
- Use **abc123** as passphrase



Steganalysis



Finally, use **stegcracker** to recover the secret file from the image.

Note: you may need to unzip the rockyou.txt wordlist file. Use the following command:

```
$ sudo gzip -d /usr/share/wordlists/rockyou.txt.gz
```

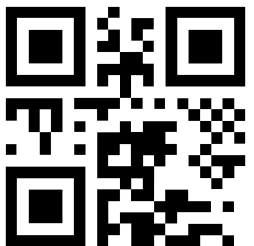


Steganography



- Can you embed a text file inside an audio/video file?
- Is there any difference between the original file and the one with your secret embedded?

Do you have any questions?



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